

DATA ENTRY ACCURACY & SPEED TRACKER

Bulk Customer Data Entry with Automated Validation & Performance Tracking
A Portfolio Project

Made by Afifah Khatoon

Acknowledgment

I would like to express my sincere gratitude to everyone who supported me in the successful completion of this project. This project was undertaken as part of my continued effort to build practical, job-ready skills in data entry, data validation, and MIS operations, alongside my B.Com studies.

I am thankful for the guidance, resources, and learning support that helped me understand real-world data entry workflows, spreadsheet-based validation techniques, and performance tracking methods used across MIS Executive and Data Entry Executive roles. This project has strengthened my practical skills in MS Excel, MS Word, and structured data handling, and has prepared me to confidently take on similar responsibilities in a professional environment.

Project Objective

The primary objective of this project is to design and build a working Excel-based tool that reflects a real MIS / Data Entry Executive workflow, with the following specific goals:

- To accurately enter bulk customer records into a structured, standardized Excel format.
- To apply data validation rules that automatically catch common data-entry errors (invalid phone numbers, malformed emails, incorrect pincodes, invalid dates), duplicate records, and more).
- To measure and demonstrate typing speed (Words Per Minute), across entry batches.
- To calculate and track data-entry accuracy on a batch-wise basis.
- To build a portfolio piece that clearly demonstrates job-readiness for Data Entry Executive, MIS Executive, and Back Office Analyst roles.

Project Overview

This project simulates a real-world bulk data entry assignment. A raw dataset of 15,000 customer records (containing intentional formatting and data-quality errors) was used as the source data. From this dataset, 500 records were manually entered into a structured Data_Entry sheet across 10 timed batches of 50 records each, replicating a typical data entry shift.

Every record entered is checked in real time against eight validation rules covering Name, Phone Number, Email, Pincode, State, Date of Registration, Amount, and Payment Mode. Records that fail any rule are automatically flagged in red with the exact field(s) at fault, while records that pass all checks are marked green. A duplicate-detection formula additionally flags any repeated Name + Phone combination.

A dedicated Speed & Accuracy Tracker sheet then pulls data live from the entry sheet to calculate Words Per Minute and Accuracy % for each batch, using the standard typing-test convention of 1 word = 5 characters. Across the full 500-record sample, the tool recorded an average typing speed of approximately 35.9 WPM and an overall accuracy of 96.8%, closely reflecting real MIS/Data Entry performance benchmarks.

The workbook is fully extensible — 200 additional blank rows are pre-loaded with the same dropdowns, validation rules, and formulas, allowing new records to be entered and validated instantly without any extra setup.

Metric	Result
Records Entered (demo batch)	500 (in 10 batches of 50)
Validation Rules Applied	8 field-level rules + duplicate check
Average Typing Speed	35.9 WPM
Overall Accuracy	96.8%
Source Dataset Size	15,000 raw customer records

Tools & Technologies Used

- MS Excel — structured data entry sheet, dropdown-based data validation, conditional formatting, formula-driven error checking (IF, AND, MATCH, COUNTIFS, SUMIF, IFERROR).
- MS Word — professional project documentation and reporting.
- Data Validation Rules — field-level checks for phone number, email, pincode, date, amount, and controlled dropdown lists for State, Product, and Payment Mode.
- Data Entry (35 WPM) — bulk record entry practiced and measured across timed batches.
- Error Tracking & Duplicate Detection — automated flagging using COUNTIFS-based logic.

Scope of the Project

This project is directly relevant to entry-level roles such as Data Entry Executive, MIS Executive, and Back Office Analyst, where speed, accuracy, and error-free bulk data handling are core requirements. Its scope includes:

- Demonstrating hands-on proficiency in structured, high-volume data entry within MS Excel.
- Showcasing the ability to design and apply data validation rules that reduce manual QC effort.
- Providing measurable, real proof of typing speed and accuracy — metrics recruiters can verify directly in the workbook.
- Serving as a reusable template that can be extended to other data types such as invoices, employee records, or inventory logs.
- Complementing existing portfolio projects (HR Analytics, Customer Behavior Analysis, Telecom Churn Prediction) by covering the operational, day-to-day MIS/data-entry skill set that analytics-only projects do not demonstrate.

Key Highlights & Core Strengths:

- Quantifiable Typing & Accuracy Benchmarks: Tracked 500 records across 10 timed batches, achieving an average typing speed of 35.9 WPM and an overall accuracy rate of 96.8%.
- Automated Error Prevention: Built 8 formula-driven validation rules (IF, AND, LEN, MATCH) and COUNTIFS-based duplicate detection to catch errors instantly in real time.
- End-to-End Tally Integration: Mapped validated customer data directly into Tally Prime Sales Vouchers with 100% data matching across customer ledgers and invoice amounts.
- Scalable Template Design: Pre-loaded 200 blank rows with dynamic formulas and data validation rules, making the sheet ready for immediate reuse in real-world workflows.

Dataset Information:

Dataset Name: Raw_Customer_Data_15000 (1).xlsx

Dataset Link: [https://github.com/afifahkhatoon61-beep/my-certificates-projects/blob/main/Raw_Customer_Data_15000%20\(1\).xlsx](https://github.com/afifahkhatoon61-beep/my-certificates-projects/blob/main/Raw_Customer_Data_15000%20(1).xlsx)

Data Entry Accuracy & Speed Tracker (Excel):

The core deliverable of this project is the Excel workbook Data_Entry_Accuracy_Speed_Tracker.xlsx, built entirely with native Excel formulas, data validation, and conditional formatting (no macros or external tools). It contains three linked sheets:

- Instructions — a legend explaining the color-coding, validation rules, and how to add new records to the sheet.
- Data_Entry — the structured entry grid where 500 customer records were entered across 10 timed batches of 50 records each.
- Speed_Accuracy_Tracker — a live dashboard that pulls data from Data_Entry to calculate typing speed and accuracy for every batch.

Work done on the Data_Entry sheet:

- Entered 500 raw customer records into a clean, standardized 18-column format (Customer ID, Name, Phone, Email, Address, City, State, Pincode, Date of Registration, Product, Amount, Payment Mode, and more).
- Built dropdown-based data validation for State, Product, and Payment Mode, so only pre-approved values can be entered — no free-text typos possible in those fields.
- Added length- and range-based validation on Phone Number (must be 10 digits), Pincode (must be 6 digits), and Amount (cannot be negative), enforced directly by Excel's Data Validation feature.
- Wrote 8 formula-driven checks (using IF, AND, ISNUMBER, MATCH, SEARCH, LEN, DATE, EOMONTH) that validate Name, Phone, Email, Pincode, State, Date, Amount, and Payment Mode for every row, and combined them into a single Validation_Status column.
- Added an Error_Details formula that names the exact field(s) that failed, so errors can be corrected in seconds instead of being re-checked manually.
- Built a duplicate-record checker using COUNTIFS on Name + Phone, flagging repeated entries automatically.
- Applied conditional formatting so passed records turn green and failed records turn red at a glance.
- Left 200 additional rows pre-loaded with the same dropdowns and formulas, ready for new records to be entered and validated instantly.

Work done on the Speed_Accuracy_Tracker sheet:

- Logged Start_Time and End_Time for each of the 10 entry batches.
- Built a Word_Count formula (LEN-based) using the standard typing-test convention of 1 word = 5 characters, applied to every field typed in each record.
- Calculated Duration (minutes) and Words Per Minute (WPM) for each batch using live formulas, not hardcoded numbers.
- Used COUNTIFS to pull the number of errors found in each batch directly from the Data_Entry sheet's Validation_Status column, then calculated Accuracy % per batch.
- Added a totals row summarizing overall performance, plus conditional formatting to highlight batches with accuracy below 90%.

Result: across the 500-record sample, the workbook shows an average typing speed of 35.9 WPM and an overall accuracy of 96.8% — both calculated live by the sheet itself, not manually entered, so the numbers update automatically if more records are added.

Data Entry Sheet — Sample View (Screenshot):

Explanation:

The screenshot below shows the Data_Entry sheet in use, with 500 customer records entered in a structured, 18-column format across 10 tracked batches. Each field — Phone Number, Email, Pincode, State, and Payment Mode — is entered directly against the dropdown lists and validation rules described in this document.

S.No	Batch_No	Customer_ID	Ful_Name	Phone_No	Email	Address	City	State	Pincode	Date_of_Regis	Product	Product_Amount	P.Payment_Mode
1	1	CUST00001	Aryan Mahara	9605012345	aryan.mahara21@gmail.com	21B, Sansi Zia	Ranchi	Jharkhand	838008	12/02/2024	File Organizer	10487.17	Net Banking
2	1	CUST00002	Ayushman Chander	9605012345	ayushman.chander10@gmail.com	890, House Circle	Arrangedad	Gauhati	815584	30/03/2023	File Organizer	7457.91	Cash
3	1	CUST00003	Thomas Sen	9696310341	thomas.sen57@gmail.com	H.No. 265, Dagar Road	Patancheru	Andhra Pradesh	164752	28/01/2025	Printer Cartridge	3509.21	Credit Card
4	1	CUST00004	Gayathri Chaudry	9657094555	gayathri.chaudry98@gmail.com	15, Lake Park	Bhopal	Bihar	855641	08/03/2024	Table Fan	10086.97	Cheque
5	1	CUST00005	Tripti Yadav	9677423884	tripti.yadav765@outlook.com	18, Patna Road	Chennai	Tamil Nadu	095132	09/11/2025	USB Hard Drive	1635.37	Cash
6	1	CUST00007	Sudisha Palla	9614892522	sudisha.palla55@outlook.com	H.No. 53, Bera Road	Jaipur	Jharkhand	818651	24/02/2024	AI Paper Ream	2897.88	Cash
7	1	CUST00008	Yash Gode	9911712227	yash.gode78@outlook.com	48, Bar Gori	Lucknow	Uttar Pradesh	248903	08/01/2026	Power Bank	11751.91	Net Banking
8	1	CUST00009	Yash Bhaty	9919713115	yash.bhaty23@outlook.com	41, Murty Chel	Mumbai	Karnataka	635901	16/07/2025	Wireless Mouse	3816.28	Cash
9	1	CUST00010	Jashu Handhava	9473829793	jashu.handhava84@yahoo.co	423, Vishanagar Circle	Chennai	Tamil Nadu	631165	17/09/2025	AI Paper Ream	7577.99	Cheque
10	1	CUST00011	Pritya Bhaty	9613337678	pritya.bhaty198@yahoo.com	287, Satya Pa	Kolkata	West Bengal	713151	14/04/2023	Laptop Stand	15564.81	Cash
11	1	CUST00012	Michael Duta	9773920606	michael.duta27@yahoo.com	18/01, Wadhwa Chowk	Patna	Bihar	486723	30/08/2024	Laptop Stand	15564.81	Net Banking
12	1	CUST00013	Ashay Contractor	9639738038	ashay.contractor85@gmail.com	511462, Rajmura Circle	Gowalpi	Assam	781381	12/02/2026	Webcam	8345.85	Cash
13	1	CUST00014	Pa Githaba	9694325346	pa.githaba12@outlook.com	49/952, Etem Park	Gauhati	Assam	781507	26/06/2026	Webcam	15937.32	Cash
14	1	CUST00015	Elvira Suresh	9251354278	elvira.suresh721@yahoo.com	H.No. 543, Khola Nagar	Gowalpi	Assam	780041	03/10/2023	Bluetooth Speaker	2022.77	Cash
15	1	CUST00016	Frederick Sander	9935349761	frederick.sander95@outlook.co	22715, Latta Street	Gowalpi	Assam	640652	28/07/2026	Bluetooth Speaker	2752.02	Cash
16	1	CUST00017	Siddharth Zacharia	9112809682	siddharth.zacharia41@gmail.c	46/578, Data Chowk	Dibrugarh	Assam	785453	29/05/2024	Whiteboard Marker S	1825.14	Net Banking
17	1	CUST00018	Pranshu Bhat	9222025953	pranshu.bhat874@gmail.com	14 No. 983, Karmam	Vaishdeva	Gujarat	160153	31/03/2025	Printer Cartridge	5021.02	UPI
18	1	CUST00019	Ekwar Bhargava	9654145686	ekwar.bhargava34@gmail.com	H.No. 051, Khama Circle	Lucknow	Uttar Pradesh	142940	11/08/2023	Webcam	7905.11	Cheque
19	1	CUST00020	Veelika Kala	9610336050	veelika.kala53@outlook.com	99/757, Ganga Marg	Contabnra	Tamil Nadu	398159	04/10/2024	Whiteboard Marker S	1299.14	Cheque
20	1	CUST00021	Gauri Sankar	9656423365	gauri.sankar784@outlook.com	67, Chanchal Nagar	Managhat	Karnataka	296458	01/01/2025	Bluetooth Speaker	4732.24	Debit Card
21	1	CUST00022	Neha Kori	9738721489	neha.kori144@gmail.com	338, Durg Chowk	Faridabad	Haryana	343320	05/04/2023	Laptop Stand	15749.19	Net Banking
22	1	CUST00023	Sudini Malik	9651672201	sudini.malik198@yahoo.com	178, Baiti Zia	Gurgaon	Bihar	338168	25/05/2024	Table Fan	1382.71	Debit Card
23	1	CUST00024	Siree Khatha	9895799897	siree.khatha118@gmail.com	77, Baiten Durg	Bilaspur	Chhattisgarh	174448	19/09/2025	Whiteboard Marker S	3019.18	Debit Card
24	1	CUST00025	Sales Register	9651672201	sales.register771@gmail.com	47, Sampal Road	Shimla	Haryana	12102325	12/02/2025	AI Paper Ream	1182.01	Debit Card
25	1	CUST00026	Mohammed Gupta	9967050668	mohammed.gupta76@outlook.c	098, Murthy Nagar	Vijayawada	Andhra Pradesh	853734	11/06/2025	USB Hard Drive	657.59	Credit Card
26	1	CUST00027	Chandreshwari	9263898901	chandreshwari.sood59@yahoo.c	H.No. 112, Bir Zia	Vijayawada	Andhra Pradesh	853734	01/11/2024	Laptop Stand	7383.25	Credit Card

Speed & Accuracy Tracker — Sample View (Screenshot):

Explanation:

The screenshot below shows the batch-wise Speed & Accuracy Tracker, calculating typing speed and accuracy live for each of the 10 entry batches based on Start_Time, End_Time, and errors flagged in the Data Entry sheet. Across all 500 records, the tracker shows an average speed of 35.9 WPM and an overall accuracy of 96.8% — both recalculated automatically as new data is entered.

	A	B	C	D	E	F	G	H	I	J
1	Speed & Accuracy Tracker									
2	Batch-wise typing speed (WPM) and data-entry accuracy, sourced live from Data_Entry sheet									
3										
4	Batch_No	Records_Enter	Start_Time	End_Time	Duration_Min	Total_Words	WPM (Words/Min)	Errors_Foun	Accuracy %	
5	1	50	09:00:00	10:41:26	41.4	1367	33.0	3	94.0%	
6	2	50	09:46:26	10:26:32	40.1	1363	34.0	3	94.0%	
7	3	50	10:31:32	11:10:36	39.1	1367	35.0	2	96.0%	
8	4	50	11:15:36	11:54:25	38.8	1358	35.0	0	100.0%	
9	5	50	11:59:25	12:37:18	37.9	1363	36.0	0	100.0%	
10	6	50	12:42:18	13:19:20	37.0	1333	36.0	1	98.0%	
11	7	50	13:24:20	14:01:03	36.7	1359	37.0	2	96.0%	
12	8	50	14:06:03	14:43:23	37.3	1381	37.0	3	94.0%	
13	9	50	14:48:23	15:24:24	36.0	1369	38.0	2	96.0%	
14	10	50	15:29:24	16:04:17	34.9	1360	39.0	0	100.0%	
15	TOTAL / AVERAGE	500				379.3	13620	35.9	16	96.8%
16	Batch_No	Data_Entry	Speed_Accuracy_Tracker							

Project Deliverable: [https://github.com/afifahkhatoon61-beep/my-certificates-projects/blob/main/Main/Data_Entry_Accuracy_Speed_Tracker%20\(1\).xlsx](https://github.com/afifahkhatoon61-beep/my-certificates-projects/blob/main/Main/Data_Entry_Accuracy_Speed_Tracker%20(1).xlsx)

Tally Voucher Mapping:

To extend the project beyond data entry and validation, a fourth sheet — Tally_Voucher_Mapping — was added to demonstrate how validated customer/invoice data can be converted into a Tally-ready Sales Voucher format.

- Built a field-mapping reference table showing how each Data_Entry column corresponds to a standard Tally voucher field — for example, Customer_ID maps to Reference No., Full_Name to Party Ledger (Debtor), Product_Purchased to Stock Item Name, Amount_Paid to the voucher Amount, and Payment_Mode to the relevant Cash/Bank ledger.
- Generated 50 sample Sales Vouchers using live formulas that pull directly from the Data_Entry sheet, auto-creating Voucher No., Voucher Date, Party Ledger Name, Sales Ledger, Stock Item, Quantity, Rate, Amount, and a system-generated Narration line for each transaction.
- This mirrors the structure Tally uses for voucher entry and data import, showing practical understanding of how day-to-day Excel/MIS data feeds into accounting software used in real back-office and MIS roles.

Excel to Tally Voucher Mapping — Sample View (Screenshot):

Explanation:

The screenshot below shows the Tally Voucher Mapping sheet, which maps each Data Entry column to its corresponding Tally voucher field (Party Ledger, Stock Item, Amount, Cash/Bank Ledger, and more). It also shows 50 sample Sales Vouchers auto-generated with live formulas from validated customer data — Voucher No., Date, Party Ledger, Stock Item, Amount, and Narration — mirroring the format Tally uses for voucher entry.

Project Deliverable: https://github.com/afifahkhatoon61-beep/my-certificates-projects/blob/main/Tally_Voucher_Mapping.xlsx

A	B	C	D	E	F	G	H	I	J	K
1	Excel to Tally Voucher Mapping									
2	Shows how mapped customer/invoice data converts into Tally-style Sales Voucher format									
3	Field Mapping Reference									
4	Data_Entry Column	Tally Voucher Field	Purpose in Tally							
5	Customer_ID	Reference No.	Unique voucher reference / cross-check with source record							
6	Ful_Name	Party Ledger	Customer ledger the sale is booked against							
7	Date of Regis/Voucher Date	Date	The Sales Voucher is generated							
8	Product_Purch Stock Item	Inventory Item	Item ledger sold in the voucher							
9	Amount_Paid	Amount	LI=Sales value entered in the voucher							
10	Payment_Mode	Cash/Bank	LI=Debit ledger where debited (Cash, Bank, UPI) account							
11	City/State	Place of Supply	Place of supply linked for GST-applicable voucher classification							
12										
13	Sample Converted Sales Vouchers (First 50 Records, generated live from Data_Entry sheet)									
14	Voucher No.	Voucher Date	Voucher Type	Party Ledger Name	Sales Ledger	Stock Item	Qty	Rate	Amount	Narration
15	SV-0001	12/02/2024	Sales	Aryan Mahara	Sales Account	File Organizer	1	10487.17	10487.17	Selling sale of File Organizer to Aryan Mahara (Ref: CUST00001) - Payment via Net Banking
16	SV-0002	30/03/2023	Sales	Ayushman Chander	Sales Account	File Organizer	1	7457.91	7457.91	Selling sale of File Organizer to Ayushman Chander (Ref: CUST00002) - Payment via Cash
17	SV-0003	28/01/2025	Sales	Thomas Sen	Sales Account	Printer Cartridge	1	3509.21	3509.21	Selling sale of Printer Cartridge to Thomas Sen (Ref: CUST00003) - Payment via Credit Card
18	SV-0004	08/03/2024	Sales	Gayathri Chaudry	Sales Account	Table Fan	1	10086.97	10086.97	Selling sale of Table Fan to Gayathri Chaudry (Ref: CUST00004) - Payment via Cheque
19	SV-0005	09/11/2025	Sales	Tripti Yadav	Sales Account	USB Hard Drive	1	1635.37	1635.37	Selling sale of USB Hard Drive to Tripti Yadav (Ref: CUST00005) - Payment via Cash
20	SV-0006	24/02/2024	Sales	Sudisha Palla	Sales Account	AI Paper Ream	1	2897.88	2897.88	Selling sale of AI Paper Ream to Sudisha Palla (Ref: CUST00006) - Payment via Cash
21	SV-0007	08/01/2026	Sales	Yash Gode	Sales Account	Power Bank	1	11751.91	11751.91	Selling sale of Power Bank to Yash Gode (Ref: CUST00007) - Payment via Net Banking
22	SV-0008	16/07/2025	Sales	Yash Bhaty	Sales Account	Wireless Mouse	1	3816.28	3816.28	Selling sale of Wireless Mouse to Yash Bhaty (Ref: CUST00008) - Payment via Cash
23	SV-0009	17/09/2025	Sales	Jashu Handhava	Sales Account	AI Paper Ream	1	7577.99	7577.99	Selling sale of AI Paper Ream to Jashu Handhava (Ref: CUST00009) - Payment via Cheque
24	SV-0010	14/04/2023	Sales	Pritya Bhaty	Sales Account	Laptop Stand	1	15564.81	15564.81	Selling sale of Laptop Stand to Pritya Bhaty (Ref: CUST00010) - Payment via Cash
25	SV-0011	30/08/2024	Sales	Michael Duta	Sales Account	Laptop Stand	1	15564.8		